

7. What are the address decoders?
8. What is a PROM?
9. What are Arrays?
10. What is Port? What are the types of Ports?

PART B — (5 × 11 = 55 marks)

Answer ALL the questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Simplify using karnaugh map
 $F(A, B, C, D) = \sum m(7, 8, 9) + d(10, 11, 12, 13, 14, 15)$.

Or

12. Convert the following numbers as stated

- (a) 4658_{10} to Binary, (3)
- (b) 110110110111_2 to Hexadecimal, (3)
- (c) 10010111_2 to decimal, (2)
- (d) $AEFB_{16}$ to octal. (3)

UNIT II

13. Explain decoders and encoders with good neat diagram.

Or

14. Explain Demultiplexers with neat diagram.

UNIT III

15. Using JK Flip Flop design a synchronous counter which counts in the sequence
000, 001, 010, 011, 100, 101, 110, 111, 000.

Or

16. Explain Ring counter.

UNIT IV

17. Explain the organization of Programmable Logic Devices (PLD).

Or

18. Implement the following function using PLA.

$$F_1 = \sum m(1, 2, 4, 6)$$

$$F_2 = \sum m(0, 1, 6, 7)$$

$$F_3 = \sum m(2, 6)$$

UNIT IV

17. Explain programmable logic arrays. (PLAs)

Or

18. Explain programmable logic devices. (PLDs)

UNIT V

19. Explain the modeling styles for combinational logic.

Or

20. Implement counter design using VERILOG HDL.

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B.Tech. DEGREE EXAMINATION,
APRIL 2015.

Third Semester

Computer Science and Engineering
DIGITAL SYSTEM DESIGN

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Convert to the decimal number 250.5 to base 3 and base 7.
2. Convert $F = xy + x'z$ to product of maxterms.
3. How many don't -case inputs are there in a BCD adder?
4. Write the truth table for full subtractor.
5. Draw the logic diagram of Master slave flip-flop.
6. Write excitation table for JK flip-flop.
7. Differentiate ROM from PLA.

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3. Implement half adder with the help of NAND gates.
4. Differentiate between multiplexer and decoder.
5. What is excitation table?
6. List out the differences between asynchronous and synchronous counters.
7. Define the term PLA.
8. What is Programmable Logic device?
9. Write the data types used in Verilog.
10. How vectors are different from arrays in Verilog?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Simplify the following Boolean function using tabulation method.

(a) $F(A, B, C, D) = \sum(2, 3, 12, 13, 14, 15)$

Or

2

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12. Simplify the following Boolean function using K-map method and implement using two-variable NAND gates
 $F(A, B, C, D, E) = \sum(0, 2, 4, 6, 9, 11, 13, 15, 17, 25, 27, 29, 31)$

UNIT II

13. Design a BCD-to-seven-segment decoder circuit.

Or

14. Implement a full-subtractor with two half-subtractors and an OR gate.

UNIT III

15. Design a counter that counts the decimal digits using 2, 4, 2, 1 code. Use T flip-flops.

Or

16. Draw the diagram of a 4-bit Johnson Counter using flip-flops.

UNIT IV

17. Derive the PLA program table for a combinational circuit that squares a 3-bit number. Minimize the number of product terms.

Or

3

6. What do you know about a ring counter? Draw a four bit ring counter.
7. What is the purpose of using decoder in a memory chip?
8. A memory stores $8K \times 16$ bytes words. How many address lines and data input / output does it have?
9. State the purpose of using HDL modeling.
10. Write a HDL program for Half Subtractor.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, ONE from each Unit.

UNIT I

11. Simplify the Boolean function using Quine — McCluskey method :

$$F = \sum(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15).$$

Or

12. Draw the NAND logic diagram that implements the following function after simplified using K map : $F(A, B, C, D) = \sum(0, 1, 2, 3, 4, 8, 9, 12).$

UNIT II

13. Explain in detail about BCD adder. design a 4 bit BCD adder.
- Or
14. (a) Implement the Boolean function $F = \sum(0, 1, 2, 5, 7, 11, 14)$ with minterms.
 - (b) Discuss in detail about parity encoder. design a parity encoder with 4 bit input.

UNIT III

15. (a) Design a synchronous counter through the sequence 2, 6, 1, 7, 5, 4. Use JK flip.
 - (b) Draw the logic diagram of a shift register with four D flipflop mux with mode selection input and explain it.
- Or
16. (a) Design a JK flip flop using D flip flop.
 - (b) Design a sequential circuit with two T flip-flops A and B and two inputs E and x. When $E = 0$, the circuit remains in its present state regardless of the value of x. When $x = 1$, the circuit goes through the following transitions from 00 to 01 to 00, and repeats. When $E = 1$, the circuit goes through the following transitions from 00 to 11 to 10 to 01 to 00 and repeats.

UNIT V

19. Elaborate the concept of 4-bit ripple carry counter with an example.

Or

20. Explain the procedure to design Multiplexer (mux 16×1) using Verilog HDL.

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2019.

Third Semester

CSE

Elective — DIGITAL SYSTEM DESIGN

(2013 – 14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Add the decimals 67 and 78 using excess-3 code.
2. Define the following : minterm and maxterm.
3. Implement half Adder using NAND Gates.
4. What is meant by self-complementing code?
5. Show the T Flipflop implementation from SR flipflop.

UNIT V

19. Explain hierarchical modeling concepts.

Or

20. Describe about design of counters and full adders.

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B.Tech DEGREE EXAMINATION, MAY 2019.

Third Semester

Computer Science Engineering

DIGITAL SYSTEM DESIGN

(2013 – 14 onwards Regulations)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Define Excess-3 code. Why is it called self-complementing code?
2. Define Demorgan's Theorems with truth table.
3. Compare half and full adders.
4. Give the differences between multiplexer and demultiplexer.
5. Define latches.
6. Give some state reduction techniques.

9. Mention the various modeling used in verilog.
10. What are the value sets in verilog?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE full question from each Unit.

All questions carry equal marks.

UNIT-I

11. Simplify the function
 $F(w,x,y,z) = \sum m(2, 3, 12, 13, 14, 15)$ using
Tabulation method. (11)

Or

12. Simplify the Boolean function to sum of product form and product of sum forms, where
 $F(A, B, C, D) = \sum m(0, 1, 2, 5, 8, 9, 10)$. (11)

UNIT-II

13. Explain the Excess 3 to BCD code converter. (11)

Or

14. How will you construct the 4×16 decoder with two 3×8 decoders. Explain the steps. (11)

UNIT-III

15. Explain the function of JK flip-flop.

Or

16. Discuss in detail about any two shift registers.

UNIT-IV

17. Explain about reduction of state and flow table with an example.

Or

18. Explain about different types of and its operation.

UNIT-V

19. Discuss the various modeling methods used in VHDL with examples.

Or

20. Write a verilog program for shift registers.

- Page 1 of 2
7. Suppose a ROM chip has 4 output lines and 9 input lines then what is the total number of bits stored in that chip?
8. What is the difference between PLA and ROM?
9. How can you declare a signal?
10. What is the use of behavioral modeling?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Using 2's complement subtract $72532 - 3250$. (3)
- (b) Using 9's complement subtract $3250 - 72532$. (3)
- (c) Simplify: $A \cdot B + \overline{A} \cdot \overline{C} + A \overline{B} C (A \cdot B + C)$. (5)

Or

12. Reduce the following function by Karnaugh map and represent the reduced function in sum of products and product of sums forms:
- $F = \pi(0, 3, 7, 8, 9, 12, 13)$.

UNIT II

13. (a) Implement a full adder in sum of products and explain.
- (b) Design a circuit for binary multiplier and explain.

Or

14. (a) Implement the following function using a 4-to-1 multiplexer:
- $F(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 15)$.

- (b) Draw a circuit for 4-bit magnitude comparator and explain.

UNIT III

15. (a) Explain the working principle of JK flip-flop.
- (b) Describe the importance of ring counter.

Or

16. (a) Construct a mod-10 counter and explain.
- (b) Write about Moore models.

7. Distinguish between volatile and non-volatile memories.
8. Write a note on static bi-polar RAM cell.
9. Give an example for Array of Instances of primitives.
10. Mention data types used in Verilog HL.

PART B — (5 × 11 = 55 marks)

Answer ALL the questions.

11. Simplify the following functions using K-map :

(a) $F(A, B, C, D) = \Sigma m(0, 3, 5, 6, 9, 10, 12, 15)$

(b) $F(W, X, Y, Z) = \Sigma m(0, 3, 4, 7, 9, 12, 14)$

Or

12. Simplify and implement the following gates using NOR gates only :

$$F = \prod M(4, 6, 10, 12, 13, 15).$$

Design and implement a full subtractor. Show how a full subtractor can be implemented using half subtractor blocks.

Or

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14. (a) Realize the function $\Sigma m(0, 3, 5, 6, 7)$ 8 : 1 multiplexer.
- (b) Elaborate on the standard IC for generator and checker.
15. Convert JK flip flop into D flip flop and T flip flop.

Or

16. Construct a shift register using SR flip flop and explain its working.
17. With the help of neat circuit diagram explain the working of MOS RAM cell.

Or

18. (a) Discuss various types of ROM.
- (b) Compare PROM, PLA and PAL.
19. Design a four bit full adder using primitive primitives and write its Verilog HDL code.

Or

3

7. What is address decoding?
8. Define. FPGA.
9. Give the uses of 4 bit ripple carry counter.
10. Write a short note on behavioral modeling.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Convert (3457.256)₈ to hexa decimal, decimal and binary. Explain. Signed binary numbers, weighted BCD codes.

Or

12. Explain K-mapping method. Reduce the following using K-mapping method

$$F = \sum m(0, 1, 2, 3, 4, 5, 7, 8, 9, 10, 12, 13)$$

Explain Error detection and correction codes.

UNIT II

13. Explain BCD adders and Code convertors.

Or

14. Explain decoder and parity encoder. Draw Circuit diagram of 3-8 line decoder.

UNIT III

15. Draw and Explain JK flip flops and conversion flip flops.

Or

16. Define. Counter.

- (a) Explain ring counter with neat diagram.
- (b) State Mealy and Moore models with necessity.

UNIT IV

17. Write a Short note on types of memories examples.

Or

18. Write Short notes on PAL, PLA and PROM.

UNIT V

19. Explain about the richness of operators available in VHDL. (11)

Or

20. Explain the following statements available in VHDL.

- (a) Process statement (5)
(b) Loop statement. (6)

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2014.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Convert $(673.124)_{10}$ into a binary number.
2. State de Morgan's law.
3. Give the characteristics of CMOS family.
4. Compare totem pole and open collector output of TTL family.
5. Implement $f = abcd$ using two input NAND Gate.
6. What is the difference between multiplexer and demultiplexer?
7. Define race around condition and explain how it can be eliminated.

7. Write the different types of ROM.
8. What is meant by race-free state?
9. Discuss the program structure of VHDL?
10. Write down the different data types in VHDL.

SECTION B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Draw the logic diagram corresponding to the following Boolean expressions without simplifying them.
 - (a) $BC' + AB + ACD$
 - (b) $(A + B)(C + D)(A' + B + D)$
 - (c) $(AB + A'B')(CD' + C'D)$
- Or
12. Express the given function in sum of min-terms and product of max-terms
 $F(A, B, C, D) = B'D + A'D + BD$

UNIT II

13. Describe the 4 bit by 3 bit binary multiplier in detail.

Or

14. Describe the logic diagram of carry look ahead generator in detail.

UNIT III

15. Describe in detail the analysis of clocked sequential circuits with D and JK flip-flops.

Or

16. Describe the shift registers and ring counter in detail.

UNIT IV

17. Explain in detail about the programmable logic array with 3 inputs, 2 outputs and 4 product terms.

Or

18. Discuss the design procedure of asynchronous sequential circuit in detail.

8. Define lockout condition.
9. What are the types of asynchronous circuit?
10. Define essential hazard.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. Draw the circuit diagram and explain the operation of two TTL NAND gate with open collector output. (11)

Or

12. Simplify the following Boolean function by using the tabulation method

$$F(w, x, y, z) = \sum(2, 3, 12, 13, 14, 15). \quad (11)$$

UNIT II

3. Design a binary to gray code converter. (11)

Or

- (a) What is a decoder? Give the truth table of 3 : 8 decoder and supplement it using basic gate. (6)

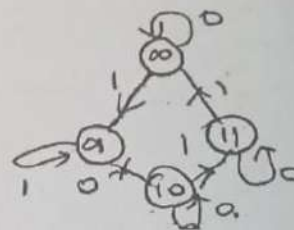
- (b) Design a 4 : 1 Mux. (5)

UNIT III

15. Design a synchronous counter for
 $4 \rightarrow 6 \rightarrow 7 \rightarrow 3 \rightarrow 1 \rightarrow 4 \dots$
 Avoid lock out condition. Use JK type design. (11)

Or

16. Design a sequential circuit for the given state diagram. (11)



UNIT IV

17. Write notes on Asynchronous and synchronous sequential circuits design. (11)

Or

18. Design a gated latch with two inputs G (gate) and D (data) and one output Q. (11)

UNIT IV

17. (a) A combinational circuit is defined by the functions: $F1 = \sum(1,3,5)$ and $F2 = \sum(5,6,7)$ implement the circuit with a PLA having 3 inputs and 3 product terms (7)
- (b) Explain the steps involved in programming of FPGA. (4)

Or

18. Implement the following using PLA.

$$A(x,y,z) = \sum(0, 1, 2, 5)$$

$$B(x,y,z) = \sum(0, 3, 5, 7)$$

$$C(x,y,z) = \sum(0, 1, 2, 3).$$

UNIT V

19. (a) Write the HDL behavioral description of a 4 to 1 line Multiplexer.
- (b) Implement full adder circuit using VHDL

Or

- (a) Realize 1X8 DEMUX using VHDL. (6)
- (b) Explain the behavioral approach for hardware description with an example (5)

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. State DeMorgan's law and prove it.
2. Express the function $F = (xy + z)(y + xz)$ in SOP form.
3. Draw the logic diagram of 2 to 4 line decoder using NAND gates.
4. What do you know about a magnitude comparator? How many outputs are there in the circuit?
5. A PN flip-flop has four operations: clear to 0, no change, complement, and set to 1, when inputs P and N are 00,01,10 and 11 respectively. Tabulate the characteristic table.

7. Compare error detection and correction.
8. What is CPLD?
9. Expand VHDL.
10. Write the types in VHDL.

PART B — (5 × 11 = 55 marks)

Answer ALL question, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the operations Boolean algebra in detail.

Or

12. How to simplify Boolean functions using Karnaugh Maps? Discuss with suitable example.

UNIT II

13. Explain the functions of adder and subtractor with logic circuits.

Or

14. Describe multiplexer and demultiplexer in detail.

UNIT III

15. What is flip-flop? Explain it with circuits.

Or

16. Discuss in detail about counters.

UNIT IV

17. Explain in detail about programmable logic array.

Or

18. Discuss in detail about reduction of state and flow tables.

UNIT V

19. Describe VHDL for combinational circuits.

Or

20. Explain VHDL for registers and counters.

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B.Tech. DEGREE EXAMINATION,
NOVEMBER 2017.

Third Semester

CSE

DIGITAL SYSTEM DESIGN

(2009-12 batches)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Convert 738 from octal to binary and decimal (738_8).
2. What is the largest hexadecimal number that can be obtained with 4 bits? What is the decimal equivalent?
3. Prove $X + X = X$.
4. Define HALF subtracter with logic diagram.
5. Define a flip flop.
6. What are counters?
7. Write short notes on RAM.

UNIT V

19. Write HDL gate level description for 3 to 8 line decoder.

Or

20. Explain how to design a multiplexor and full adder using Verlog HDL?
-

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2017.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Convert (101101.1101) to decimal and hexadecimal form.
2. What do you know about Universal Gates?
3. Draw the schematic of Half Subtractor logic.
4. Distinguish between a decoder and a demultiplexor?
5. What is race around condition?
6. What is a Mealy Circuit?

8. What is FPGA?
9. What are registers?
10. Define Behavioral models.

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

11. Perform the subtraction with the following decimal numbers using 10's and 9's complement.

- (a) 5250 – 321
- (b) 753 – 864
- (c) 3570 – 2100
- (d) 20 – 1000.

Or

12. Solve by tabulation method.

- (a) $F(A, B, C, D) = (0, 1, 2, 3, 4, 10, 11)$
- (b) $F(A, B, C, D) = (2, 3, 12, 13, 14, 15)$
- (c) $F = ACE + A'CD'E' + A'C'DE$ with DON'T care $d = DE' + A'D'E + AD'E'$.

13. Explain in detail about Binary parallel adder logical diagram.

Or

14. Simplify the following expressions and implement them using NAND.

- (a) $AB' + ABD + ABD' + A'C'D' + A'BC'$
- (b) $BD + BCD' + AB'C'D'$

15. Design a BCD counter using JK flip flop.

Or

16. Design a counter with the following sequence 0, 1, 3, 2, 6, 4, 5, 7 and repeat. Use RS flip flop.

17. Write notes on virtual memory.

Or

18. Explain in detail about mass storage devices.

19. Write in detail the design flow and structure of VHDL.

Or

20. Explain any two combinational circuits.

UNIT V

19. Explain and design a 4-bit ripple carry counter.
Or
20. Explain Gate-level and Dataflow Modeling.
-

Find the class label for the

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2015.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

(2013 – 2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What are Gray Codes?
2. What are the advantages of Quine-McCluskey method?
3. Write the sum and carry equation of full adder.
4. What are demultiplexers?
5. Write the properties of D Flip Flops.
6. What are the advantages of Moore model?

6. When is a counter said to suffer from lockout?
7. Distinguish between EPROM and EEPROM.
8. Where do we use PLA's?
9. How the numbers are specified in Verilog?
10. Define the term Gate level.

PART B — (5 × 11 = 55 marks)

Answer ALL question, ONE from each Units.

All question carry equal marks.

UNIT I

11. Solve the following using K-map and verify by using Boolean algebra :

$$F(W, X, Y, Z) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$$

Or

12. Minimize the expression using Quine Mccluskey (tabulation) method

$$F(W, X, Y, Z) = \sum m(0, 3, 5, 6, 7, 10, 12, 13) + \sum d(2, 9, 15)$$

UNIT II

13. Solve a combinational circuit to convert 8421 code to Excess 3 code.

Or

14. (a) Design full adder using NAND gates.
(b) Design a 4-bit 2's complement circuit.

UNIT III

15. Design a three bit synchronous counter with flop and draw the circuit diagram.

Or

16. Design a synchronous counter which count sequence $0 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 7 \rightarrow 0$, flip-flop.

UNIT IV

17. Deduce the PLA for the following :

$$F1 = (a, b, c) = \sum m(0, 1, 2, 4)$$

$$F2 = (a, b, c) = \sum m(0, 5, 6, 7)$$

Or

18. Illustrate with neat sketch and categories of RAM.

UNIT IV

17. Design a control unit with PLA and demonstrate the procedure with an example.

Or

18. Explain the internal construction of field programmable gate array.

UNIT V

19. (a) Write VHDL code for a full subtractor using logic equations. (5)
(b) Explain how the procedure is declared in VHDL. (6)

Or

Explain the VHDL model for a counter and write a VHDL description of the counter.

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2014.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. State the distributive law of boolean algebra.
2. Expand the expression $X + \bar{Y}$ in minterms.
3. What is decoder?
4. Give the boolean functions for full-subtractor.
5. What are the two types of sequential circuits?
6. What is hold time?

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B.Tech. DEGREE EXAMINATION, MAY 2019.

Third Semester

Computer Science and Engineering/Information
Technology

DIGITAL SYSTEM DESIGN

(2003-08 and 2009 - 12 Regulations)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Which gates are called as universal gates? What are its advantages?
2. Reduce the boolean expression $A(A+B)$.
3. Write the logic equation for half adder.
4. Why parity checker is needed?
5. Give a brief note on latch.
6. What is Thonson counter?
7. Write the significance of state assignment.
8. List the types of ROM.

UNIT V

Describe the VHDL for combinational circuits in detail.

Or

Explain the VHDL for binary multiplier in detail.

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2014.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Time : Three hours

Maximum : 75 marks

SECTION A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Convert the binary number 0.0110 into decimal.
2. Simplify the given Boolean functions to a minimum number of literals $(x + y)(x + y')$.
3. Define: Decoder.
4. Write short note on Multiplexers.
5. Discuss briefly about D-latch.
6. List the features of shift registers.

19. Explain 4 bit ripple carry counter.

Or

20. Write a note on :

- (a) Gate level Modeling
- (b) Dataflow Modeling.

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B.Tech. DEGREE EXAMINATION, MAY 2018.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

(2013 – 2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. State the distributive property of Boolean algebra.
2. Show that $(X + Y' + XY)(X + Y)(XY) = 0$.
3. Define Combinational circuit.
4. List out the applications of decoder.
5. What is a Counter?
6. Differentiate combinational circuits and sequential circuits.
7. Mention two differences between PLD and ROM.

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2016.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

(2013-2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Find the octal equivalent of decimal 200.
2. State De Morgan's theorem.
3. What is meant by decoder?
4. What is multiplexer?
5. What is ring counter?
6. What is JK flip-flop?
7. Distinguish between RAM and ROM.

9. List the levels of abstraction in Verilog HDL design.
10. Define Instances.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Explain Gray Code and list them. (5)
- (b) Explain in detail about error detection and correction codes. (6)

Or

12. Express the Boolean function $F = A + B'C$ as a sum of minterms. (11)

UNIT II

13. Explain the design principles of Full Adder circuit using combinational logic, show the K-Maps and Implementation of full adder in sum-of-products form. (11)

Or

14. Explain the design principles of Four-to-one-line multiplexer and give an example of Boolean function implementation. (11)

2

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UNIT III

15. Explain the working principles of JK and T flop with Characteristic Tables and Characteristic Equations.

Or

16. Explain in detail about the Four-bit universal shift register.

UNIT IV

17. Explain in detail about the organization of ROM.

Or

18. Explain the design and implementation of Programmable Array Logic (PLA).

UNIT V

19. Explain the Components of a Verilog HDL Simulation with an example.

Or

20. Design and implement a 4 bit ripple carry counter using Verilog HDL.

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UNIT V

19. Describe the VHDL for registers and counters in detail.

Or

20. Explain the VHDL for binary multiplier in detail.

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2016.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

(2009-2012 batches)

Time : Three hours

Maximum : 75 marks

SECTION A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

1. Using 10's complement, subtract 3250-72532.
2. What is meant by don't care conditions?
3. Discuss briefly about subtractors with circuit diagram.
4. Draw the circuit diagram for 4-bit adder.
5. Discuss the block diagram of sequential circuit
6. Write note on BCD ripple counter.

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B.Tech. DEGREE EXAMINATION,
NOVEMBER 2017.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

(2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Convert decimal 41 to binary.
2. Define Duality in Boolean algebra.
3. Define Carry Propagation.
4. What is Three-State Gate?
5. Define Latch.
6. What is state diagram?
7. Define memory *refreshing*.
8. What is *Hamming Code*?

8. Differentiate between PLA and PAL.
9. Define modules in verilog.
10. What is meant by full adder?

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. Discuss the boolean algebra and theorem with example.

Or

12. Elucidate the minimization of expression using Karnaugh maps and Quine-McCluskey.

UNIT II

13. Explain half adder and full adder with a neat circuit diagram.

Or

14. Illustrate multiplexer and de-multiplexer with a neat circuit diagram.

UNIT III

15. Explain the edge-triggered JK and D flip flops.

Or

16. Design GCD processor with necessary state table and state reduction techniques.

UNIT IV

17. Explain how the RAM and ROM chips are organized and how the addresses are decoded to select those chips.

Or

18. Illustrate the combinational logic implementation using PROMs, PLAs, PALS.

UNIT V

19. Elucidate the 4-bit ripple carry counter with a neat diagram.

Or

20. Discuss the data types used in Verilog.

7. Determine the no of address lines required for 512 bytes of memory and for 2KB memory.
8. Why we need programmable Logic Devices?
9. Why we need Verilog HDL?
10. List the modeling techniques available in HDL.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Convert $(1947)_{10}$ into its equivalent Octal and Decimal representation (7)
 - (b) Perform $(147 - 89)$ using 2's complement binary arithmetic. (4)
- Or
12. Simplify the following Boolean function F using Karnaugh map method:

- (a) $F(A, B, C, D) = \Sigma (1, 4, 5, 6, 12, 14, 15)$ (2)
- (b) $F(A, B, C, D) = \Sigma (0, 1, 2, 4, 5, 7, 11, 15)$ (3)
- (c) $F(A, B, C, D) = \Sigma (2, 3, 10, 11, 12, 13, 14, 15)$ (3)
- (d) $F(A, B, C, D) = \Sigma (0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$ (3)

UNIT II

13. Explain the design a Full Adder and Subtractor using truth table and Logic circuits.
- Or
14. With suitable example explain how to implement the combinational circuits with multiplexers.

UNIT III

15. Describe the steps for design of asynchronous sequential circuit.
- Or
16. Design a shift register using JK flip flop.

UNIT IV

17. (a) Write short notes on the basic configurations of the 3 types of programmable logic devices.
 - (b) Draw the signals of a 32×8 RAM with control input.
Show the external connections necessary to have a 128×8 RAM using a decoder and replication of this RAM.
- Or
18. Explain the concept of Field Programmable Gate Arrays.

7. Define memory buffer register.
8. Differentiate synchronous and asynchronous sequential circuit.
9. List the features of VHDL.
10. List VHDL design units to describe an entity.

PART B — (5 × 11 = 55 marks)

Answer ALL the questions.

11. Explain the following functions in sum of minterms and a product of maxterms.

(a) $F(A, B, C, D) = (A + B' + C)(A + B')$
 $(A + C' + D')(A' + B + C + D')(B + C' + D')$ (5½)

(b) $F(a, b, c, d) = c'd + abc + abd' + a'b'd$ (5½)

Or

Simplify each of the following function and implement them with NOR gates. Give two alternatives.

(a) $F_1 = AC' + ACE + ACE' + A'CD' + A'D'E'$ (5½)

(b) $F_2 = (B' + D')(A' + C + D)(A + B' + C' + D)$
 $(A' + B + C' + D')$ (5½)

13. (a) Design a combinational circuit with 4 input lines that represent a decimal digit in BCD and four output lines that generate 9's complement of the input digit.
- (b) Show how a full-adder can be converted to full-subtractor with the addition of an inverter circuit.

Or

14. (a) Design a BCD-to-excess 3 code converter with a BCD-to-decimal decoder and four gates.
- (b) Draw the logic diagram of 2-line to 4-line decoder and demultiplexer using NOR gates only.
15. (a) Design BCD counter with JK flip-flop.
- (b) Design a counter with the following binary sequence: 0, 1, 3, 7, 6, 4 and repeat. Use T flip-flop.

Or

16. (a) Draw the logic diagram of a master-slave flip-flop. Use NAND gates.
- (b) Explain binary up-down counter.

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Convert the 364 decimal into BCD equivalent.
2. State error detecting and correction codes.
3. Convert Gray number 1110 to its BCD equivalent.
4. Annotate encoder.
5. Define counters.
6. Annotate the shift register
7. List out the types of memories.

17. How the information is stored in magnetic core memory? Explain it. (11)

Or

18. Explain race free state assignment with 3-row, 4-row and multiple row method. (11)

19. Explain various ways to describe the sequential circuit in VHDL. (11)

Or

- Write an algorithm for 4×4 unsigned multiplier. Write VHDL code for it. (11)

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B.Tech DEGREE EXAMINATION,
MAY 2018.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

(2009-12 Batches)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Define reflected code.
2. Find 10's complement of $(853)_{10}$.
3. Differentiate combinational and sequential circuit.
4. Draw the implementation of $F = A(B+CD) + B\overline{C}$ with NAND gates.
5. Define clocked sequential circuit.
6. Mention the function of ripple counter.

8. Determine the number of inputs, the number of outputs, and the type of output for each of the following PAL part number

(a) PAL 12H6

(b) PAL 16L2.

9. What are directives?

10. Define behavioral Modelling.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE questions from each Unit.

All questions carry equal marks.

11. Minimize the expression using Quine McCluskey (Tabulation) method

$$F = \sum m(0, 1, 9, 15, 24, 29, 30) + \sum d(8, 11, 31)$$

Or

12. Simplify the following boolean expression in

(a) Sum of Product

(b) Product of Sum using Karnaugh Map

$$AC + BD + A'CD + ABCD$$

13. What is a decoder? Construct a 5 to 32 decoder using one 2 to 4 decoder and four 3 to 8 decoders.

Or

14. Realize the Sum of Products expression $Y = \sum M(0, 5, 10, 15)$ using 4:1 Multiplexers.

15. Write the excitation table of JK-flip flop. Explain JK Flip Flop.

Or

16. Design a synchronous counter which counts 001, 010, 011, 100, 101, 110, 111, 000 using D-FF.

17. How does a PLA differ from PAL? Implement the following functions using ROM.

$$W(A, B, C, D) = \sum m(3, 7, 8, 9, 11, 15)$$

$$X(A, B, C, D) = \sum m(3, 4, 5, 7, 10, 14, 15)$$

$$Y(A, B, C, D) = \sum m(1, 5, 7, 11, 15)$$

Or

18. Draw a basic logic diagram for a 256 × 8 bit RAM, showing all the inputs and outputs. Mention the capacity of a DRAM that has address lines.

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B.Tech. DEGREE EXAMINATION, APRIL 2017.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

(2009–12 Batches)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Convert the decimal number 524 into binary.
2. What are binary codes?
3. Write the function of encoder.
4. Comment on parity code.
5. What are latches?
6. Write the use of registers.

7. Write the hamming code for the original data 1101.
8. What is meant by coincident decoding?
9. Discuss the program structure of VHDL.
10. Write the program to add two four bit vectors using function written in package "bit_pack".

SECTION B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Draw the logic diagram corresponding to the following Boolean expressions without simplifying them.
 - (a) $BC' + AB + ACD$
 - (b) $(A+B)(C+D)(A'+B+D)$
 - (c) $(AB+A'B')(CD'+C'D)$

Or

12. Simplify the given Boolean function in product of sums.

$$F(A, B, C, D) = \Sigma(0, 1, 2, 5, 6, 9, 10).$$

UNIT II

13. Describe the binary parallel adder and decoder adder in detail.

Or

14. Describe the 4-bit magnitude comparator in detail.

UNIT III

15. Describe in detail the analysis of combinational sequential circuits with D and JK flip-flops.

Or

16. Describe the ripple counters and synchronous counters in detail.

UNIT IV

17. Explain the procedure for implementation of asynchronous sequential circuits using SR latches.

Or

18. Discuss the procedure for reduction of state flow tables in asynchronous sequential circuit.

6. What do you know about a ring counter? Draw a four bit ring counter.
7. What is the purpose of using decoder in a memory chip?
8. A memory stores $8K \times 16$ bytes words. How many address lines and data input / output does it have?
9. State the purpose of using HDL modeling.
10. Write a HDL program for Half Subtractor.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, ONE from each Unit.

UNIT I

11. Simplify the Boolean function using Quine — McCluskey method :
 $F = \sum(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$.
- Or
12. Draw the NAND logic diagram that implements the following function after simplified using K map : $F(A, B, C, D) = \sum(0, 1, 2, 3, 4, 8, 9, 12)$.

UNIT II

13. Explain in detail about BCD adder circuit. design a 4 bit BCD adder.
- Or
14. (a) Implement the Boolean function $F(A, B, C) = \sum(0, 1, 2, 5, 7, 11, 14)$ with multiplexer.
(b) Discuss in detail about parity encoder. Design a parity encoder with 4 bit input.

UNIT III

15. (a) Design a synchronous counter that through the sequence 2, 6, 1, 7, 5, 4 and repeats. Use JK flip.
(b) Draw the logic diagram of a 4-bit universal shift register with four D flipflop and four mux with mode selection input's S1 and S0 and explain it.
- Or
16. (a) Design a JK flip flop using D flip flops.
(b) Design a sequential circuit with two T flip-flops A and B and two inputs E and x. When $E = 0$, the circuit remains in the same state regardless of the value of x. When $E = 1$, the circuit goes through the state transitions from 00 to 01 to 10 to 11 to 00, and repeats. When $E = 1$ and $x = 1$, the circuit goes through the state transitions from 00 to 11 to 10 to 01 back to 00 and repeats.

To
 Mrs. CELIN JULIE R
 94, Kamaraj
 Puducherry

8. Do's and Don'ts of FPGA's.
9. Define Behavioral models.
10. What are the datatypes allowed in VERILOG.HDL?

PART B — (5 × 11 = 55 marks)

Answer ALL questions one from each unit.

All questions carry equal marks.

UNIT I

11. Perform subtraction with the following decimal numbers using 10's complement and 9's complement.
 - (a) 5250-321
 - (b) 3570-2100
 - (c) 753-864
 - (d) 20-1000.

Or

12. Simplify the following boolean function using Quine-Mcclusky method.

$$F = \Sigma(0,2,4,6,8,9,10,11,12,14)$$

UNIT II

13. Explain the working principle of half adder and full adder.

Or

14. Design a combinational circuit with a decoder and external gates using the following function.

$$F_1 = x'y' + xyz'$$

$$F_2 = x' + y$$

$$F_3 = xy + x'y'$$

UNIT III

15. Design a counter with the following binary sequence: 0,1,3,2,6,4,5,7 and repeat. Use RS flip-flop.

Or

16. A sequential circuit has 2 flip-flop, 2 inputs and 2 outputs. The flip-flop input function and output function are as follows:

$$JA = xB + y'B'$$

$$JB = xA'$$

$$Z = xyA + x'y'B$$

$$KA = xy'B'$$

$$KB = xy' + A$$

Obtain the logic diagram, state table, state diagram and state equation.

18. Prove that a 32×8 ROM can be used to implement a circuit that generate the binary square of an 5-input number with $B_0 = A_0$ and $B_1 = 0$. Draw a block diagram of the circuit and list the first four and the last four entries of the ROM truth table.

UNIT V

19. Explain the modules involved in Verilog.

Or

20. Define a 4-to-1 multiplexer with the behavioral case statement using Verilog HDL.

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2018.

Third Semester

Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Perform the subtraction with following decimal numbers using
 - (a) 10's complement and
 - (b) 9's complement. Check the answer by straight subtraction.
 - (i) $5250 - 321$
 - (ii) $20 - 1000$
2. Convert the following to the other canonical form

$$F(x, y, z) = \sum (1, 3, 7)$$

8. What is meant by address decoding?
9. What is meant by gate-level modeling?
10. What is meant by dataflow modeling?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Using the K-map method, obtain the minimal sum of product expression of the following function
 $y = \Sigma(0, 2, 3, 6, 7, 8, 10, 11, 12, 15)$.

Or

12. Find the minimal sum of products for the Boolean expression

$$f(w, x, y, z) = \Sigma(1, 3, 4, 5, 9, 10, 11) + \Sigma\phi(6, 8).$$

Using the Quine-McClusky method. And also draw the logic circuit diagram.

UNIT II

13. Design a full subtractor using only NOR gates.

Or

14. Briefly discuss about parallel adders.

UNIT III

15. Explain about different types of Shift Registers in detail.

Or

16. Discuss about mealy and Moore models in detail.

UNIT IV

17. Briefly discuss about Programmable Array Logic (PAL) devices.

Or

18. Explain about different types of memories in detail.

UNIT V

19. Write notes on hierarchical Modeling concepts.

Or

20. Explain about 4-bit ripple carry counter in detail.